

ANIMAL BEHAVIOUR

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ALL ANIMALS RESPOND to their surroundings. A cat, for example, will arch its back when threatening a rival, but lower its body when stalking a mouse. Everything that an animal does, and the way in which it does it, makes up its behaviour. An animal's behaviour enables it to increase its chances of survival and find a mate so that it can pass on its genes to the next generation. Some behaviours are inbuilt, or instinctive; others are learned during the animal's lifetime.

Instinctive behaviour

Instinct is a term used to describe behaviours that an animal performs automatically without having to learn them. Instinctive behaviour is programmed by an animal's genes. It consists of unchanging components called fixed-action patterns. The fixed-action pattern often begins when an animal responds to a feature in its surroundings or on another animal, called a sign stimulus.

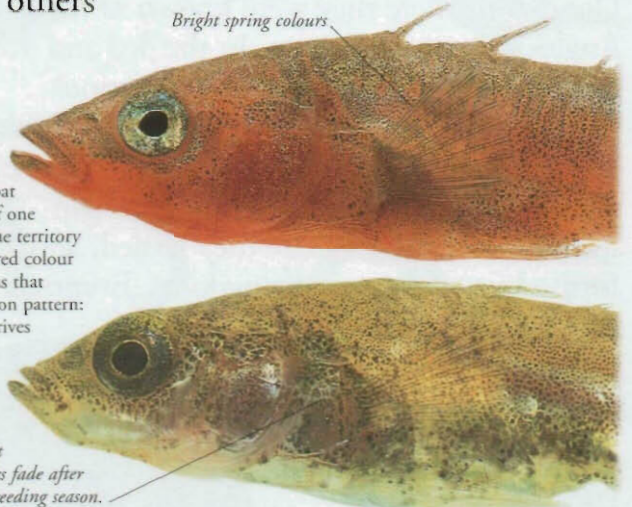


Web spinning

Many species of spider, including this black widow spider, spin webs in order to trap their insect food. Web spinning is purely instinctive. A spider would not have time in its limited life to learn how to construct such a complex structure.

Sign stimulus

In the spring, when these freshwater fish breed, the male's throat and belly turn red. If one male intrudes into the territory of another male, its red colour acts as a sign stimulus that produces a fixed-action pattern: the occupying fish drives out the intruder.



Egg-rolling

Greylag geese nest on the ground. If an egg rolls out of the nest, the female goose automatically reaches out with her neck and pulls the egg back in. By being in the wrong place, the egg acts as a sign stimulus that causes the female to carry out the fixed-action pattern of egg-rolling.

Bright spring colours

Bright colours fade after the breeding season.

Learned behaviour

Learning occurs when an animal adapts to its surroundings by changing its behaviour. By responding to experiences and adapting to changing conditions, an animal increases its chances of survival. Learning takes time, and animals that are dependent on learned behaviour have long lives and large brains.

Trial and error learning

An animal will associate an action it carries out with a successful result, such as getting food or defeating a rival. This "reward" motivates the animal to alter its behaviour to improve the result of future actions.

Puppies play-fight and perfect their hunting skills.



Young ducklings follow their mother.



Imprinting

This is shown by some young animals that make a strong bond with their parent soon after hatching or birth. Young ducklings, for example, stay close to their mother and improve their chances of survival under her protection.



Learning tool use

Some animals learn to use simple "tools" in order to feed. Sea otters, found off the coast of California, USA, swim on their backs with a stone on their chests on which they smash the shells of clams and mussels to get at the juicy contents. Young otters learn tool use from their parents.



Insight learning

This involves a form of reasoning. Some animals can solve new problems by drawing on past experiences. Chimpanzees, having learned to extract termites or ants from a nest with a stick, can exploit any shape or size of nest.

Communication

Animals communicate by sending out signals that are recognized by other animals and alter their behaviour in some way. The signals can be sights, sounds, or scents. Communication is used, for example, to find a mate, threaten rivals or enemies, defend a territory, warn of danger, or hold a group together.

Visual signals

Animals may use visual signals as a threat or to attract a mate. This puss moth caterpillar adopts a warning posture if threatened by an enemy. An enemy that ignores the warning is rewarded with a stinging squirt of formic acid.



Bright colours add to the warning.

Puss moth caterpillar



Song thrush sings from a perch.

Sound

Many animals, including crickets, bullfrogs, peacocks, and whales, use sound to communicate. This male song thrush sings to proclaim his territory, to warn rivals to stay away, and to attract a female.

Chemicals

Some animals release chemicals called pheromones, which, when detected, affect the behaviour of other members of the same species. Female gypsy moths release pheromones that attract males from several kilometres away.



Gypsy moth